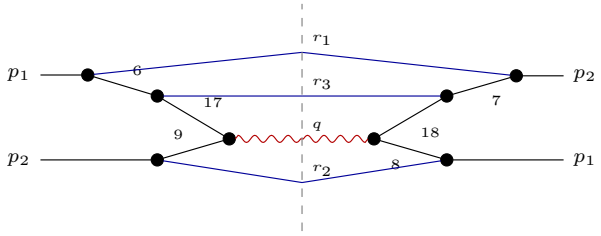
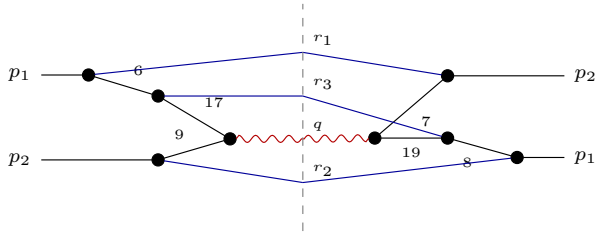
Integral families as cut diagrams

Amplitude (left) × conjugate (right) for $p_1 p_2 \to V(q)\, r_1 r_2 r_3$, glued along the cut (dashed): r_1, r_2, r_3 (blue), q (red, $q^2 = m^2$). The rapidity condition $-q \cdot p_1 + u\, q \cdot p_2 = 0$ of every family is not a line. Internal lines carry their DD number. **Regular propagators** (DD numbers, all legs incoming): $6 = (p_1 + r_1)^2$; $7 = (p_2 + r_1)^2$; $8 = (p_1 + r_2)^2$; $9 = (p_2 + r_2)^2$; $10 = (p_1 + r_3)^2$; $11 = (p_2 + r_3)^2$; $12 = (r_1 + r_2)^2$; $13 = (r_2 + r_3)^2$; $14 = (r_1 + r_3)^2$; $15 = (p_1 + r_1 + r_2)^2$; $16 = (p_2 + r_1 + r_2)^2$; $17 = (p_1 + r_1 + r_3)^2$; $18 = (p_2 + r_1 + r_3)^2$; $19 = (p_1 + r_2 + r_3)^2$; $20 = (p_2 + r_2 + r_3)^2$; $21 = (r_1 + r_2 + r_3)^2$; $22 = (p_1 + r_1 + r_2 + r_3)^2$; $23 = (p_2 + r_1 + r_2 + r_3)^2$; $24 = (p_1 + p_2 + r_1)^2$; $25 = (p_1 + p_2 + r_2)^2$; $26 = (p_1 + p_2 + r_3)^2$; $27 = (p_1 + p_2 + r_1 + r_2)^2$; $28 = (p_1 + p_2 + r_1 + r_3)^2$; $29 = (p_1 + p_2 + r_2 + r_3)^2$.



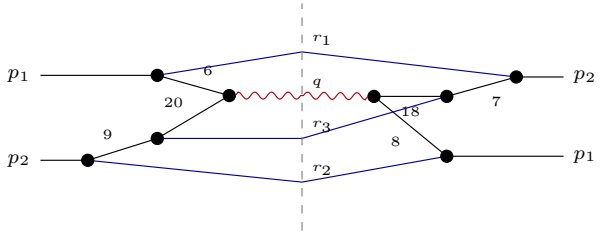
FAM[1] (topo1)

left: 6, 9, 17; right: 7, 8, 18; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



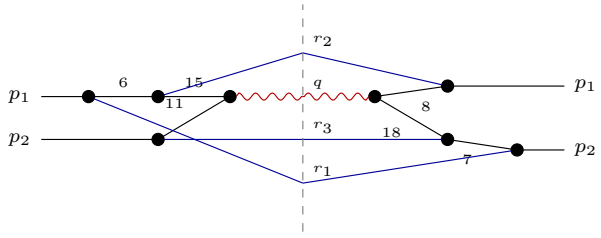
FAM[7] (topo7)

left: 6, 9, 17; right: 7, 8, 19; numerators: $13 = (r_2 + r_3)^2$



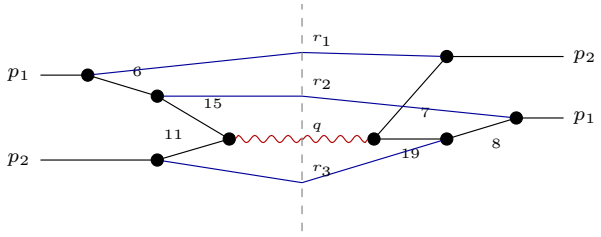
FAM[13] (topo13)

left: 6, 9, 20; right: 7, 8, 18; numerators: $13 = (r_2 + r_3)^2$



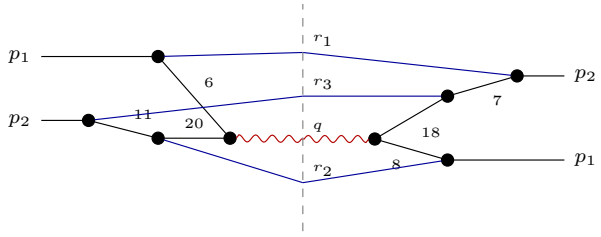
FAM[19] (topo19)

left: 6, 11, 15; right: 7, 8, 18; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



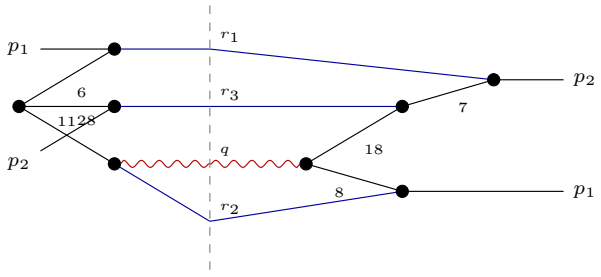
FAM[25] (topo25)

left: 6, 11, 15; right: 7, 8, 19; numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



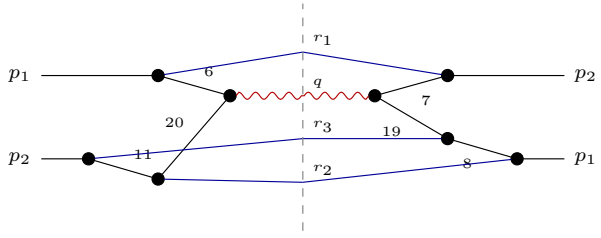
FAM[31] (topo31)

left: 6, 11, 20; right: 7, 8, 18; numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



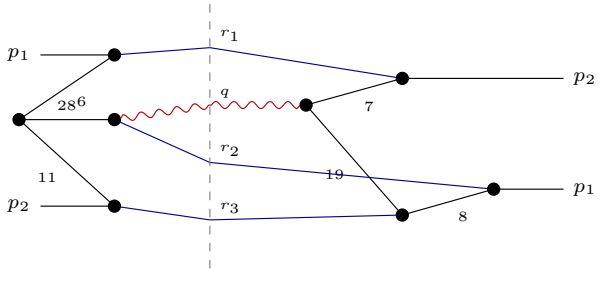
FAM[37] (topo37)

left: 6, 11, 28; right: 7, 8, 18; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



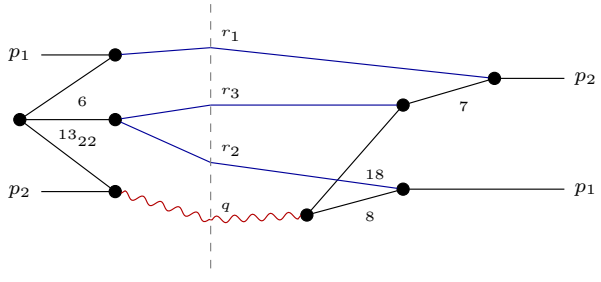
FAM[43] (topo43)

left: 6, 11, 20; right: 7, 8, 19; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



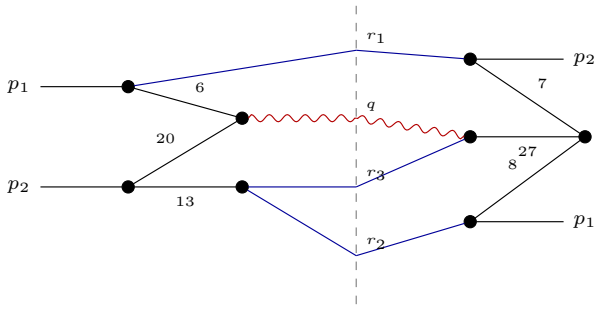
FAM[49] (topo49)

left: 6, 11, 28; right: 7, 8, 19; numerators: $13 = (r_2 + r_3)^2$



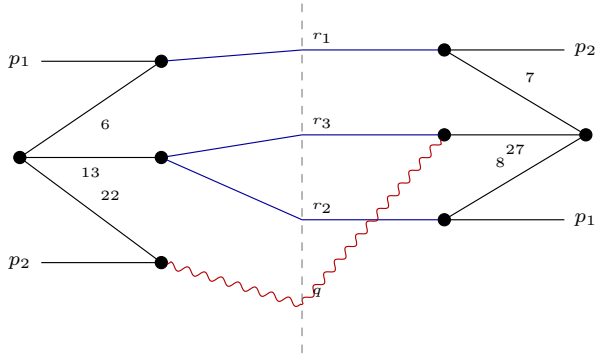
FAM[55] (topo55)

left: 6, 13, 22; right: 7, 8, 18; numerators: $14 = (r_1 + r_3)^2$



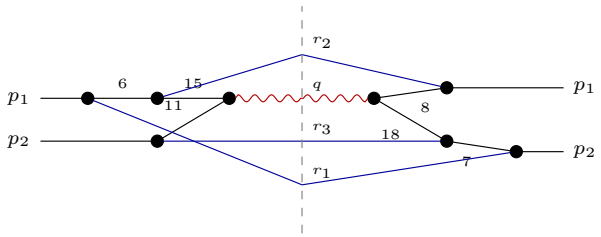
FAM[61] (topo61)

left: 6, 13, 20; right: 7, 8, 27; numerators: $25 = (p_1 + p_2 + r_2)^2$



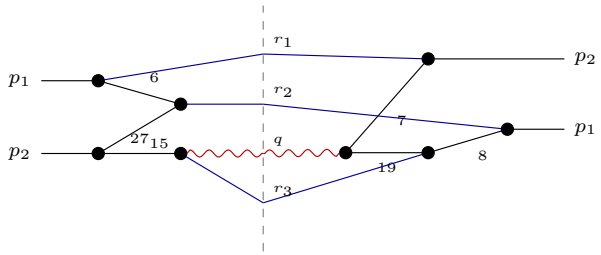
FAM[67] (topo67)

left: 6, 13, 22; right: 7, 8, 27; numerators: $15 = (p_1 + r_1 + r_2)^2$



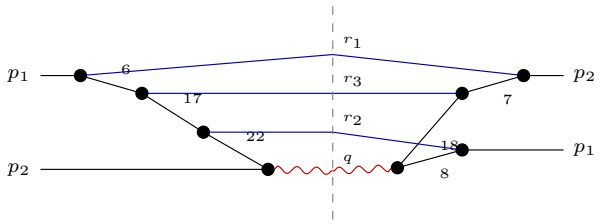
FAM[73] (topo73)

left: 6, 11, 15; right: 7, 8, 18; numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



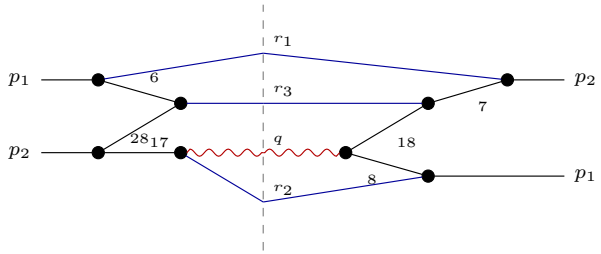
FAM[79] (topo79)

left: 6, 15, 27; right: 7, 8, 19; numerators: $13 = (r_2 + r_3)^2$



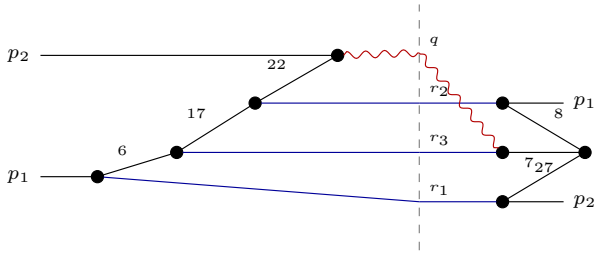
FAM[85] (topo85)

left: 6, 17, 22; right: 7, 8, 18; numerators: $19 = (p_1 + r_2 + r_3)^2$



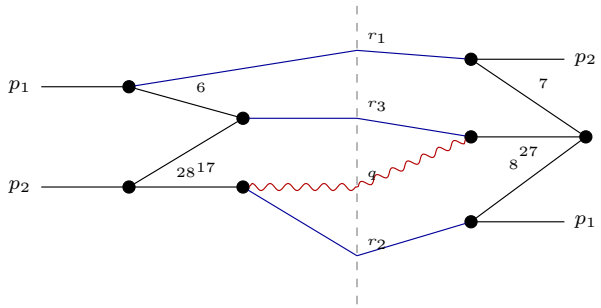
FAM[91] (topo91)

left: 6, 17, 28; right: 7, 8, 18; numerators: $19 = (p_1 + r_2 + r_3)^2$



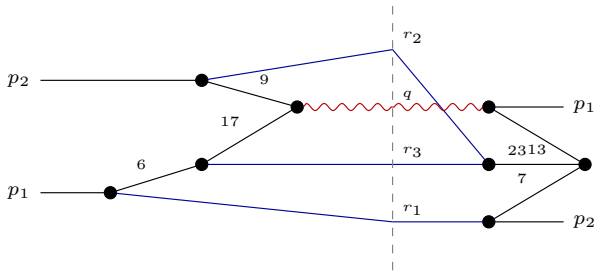
FAM[97] (topo97)

left: 6, 17, 22; right: 7, 8, 27; numerators: $25 = (p_1 + p_2 + r_2)^2$



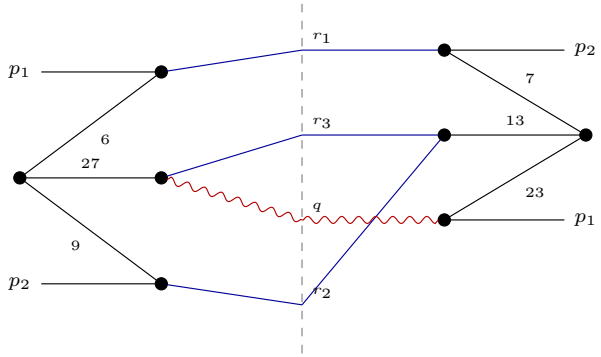
FAM[103] (topo103)

left: 6, 17, 28; right: 7, 8, 27; numerators: $14 = (r_1 + r_3)^2$



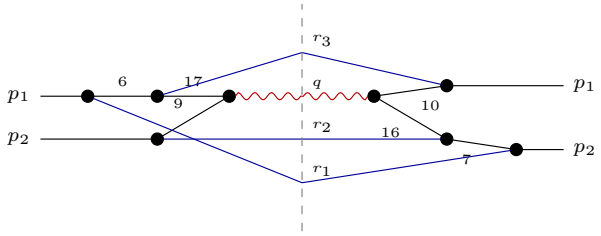
FAM[109] (topo109)

left: 6, 9, 17; right: 7, 13, 23; numerators: $14 = (r_1 + r_3)^2$

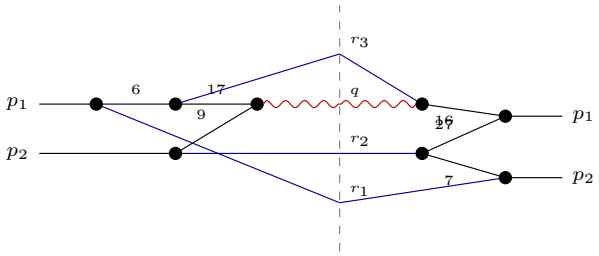


FAM[115] (topo115)

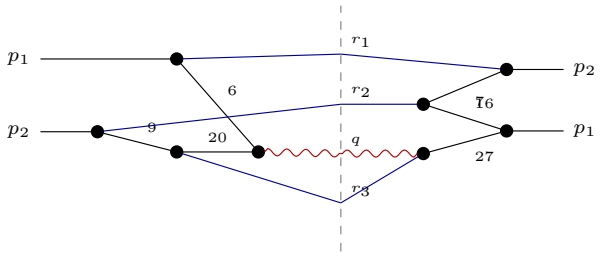
left: 6, 9, 27; right: 7, 13, 23; numerators: $16 = (p_2 + r_1 + r_2)^2$



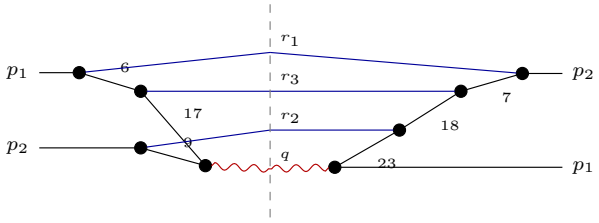
FAM[121] (topo121)
left: 6, 9, 17; right: 7, 10, 16; numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



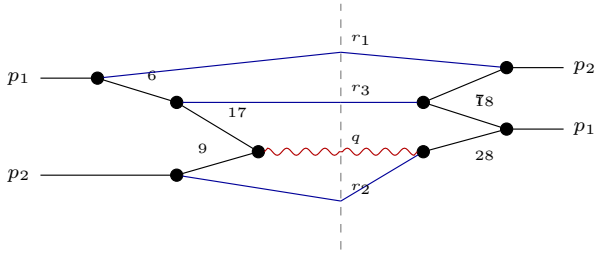
FAM[127] (topo127)
left: 6, 9, 17; right: 7, 16, 27; numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



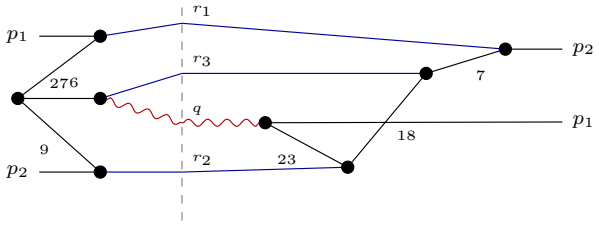
FAM[133] (topo133)
left: 6, 9, 20; right: 7, 16, 27; numerators: $13 = (r_2 + r_3)^2$



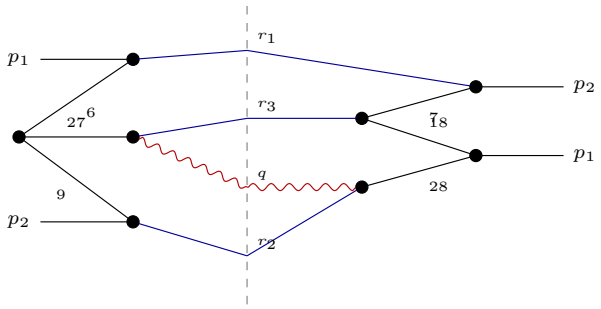
FAM[139] (topo139)
left: 6, 9, 17; right: 7, 18, 23; numerators: $20 = (p_2 + r_2 + r_3)^2$



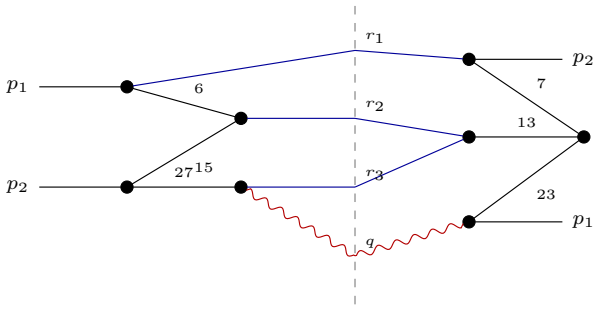
FAM[145] (topo145)
left: 6, 9, 17; right: 7, 18, 28; numerators: $20 = (p_2 + r_2 + r_3)^2$



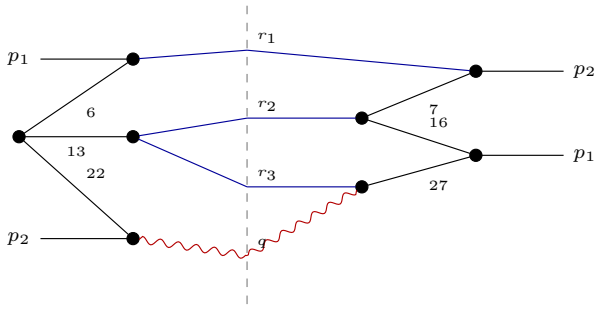
FAM[151] (topo151)
left: 6, 9, 27; right: 7, 18, 23; numerators: $25 = (p_1 + p_2 + r_2)^2$



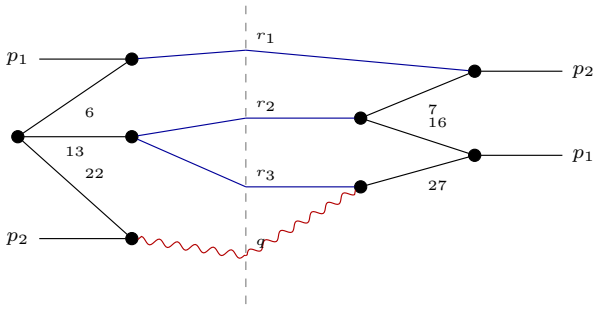
FAM[157] (topo157)
left: 6, 9, 27; right: 7, 18, 28; numerators: $14 = (r_1 + r_3)^2$



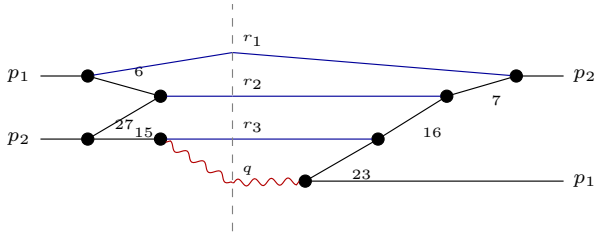
FAM[163] (topo163)
left: 6, 15, 27; right: 7, 13, 23; numerators: $12 = (r_1 + r_2)^2$



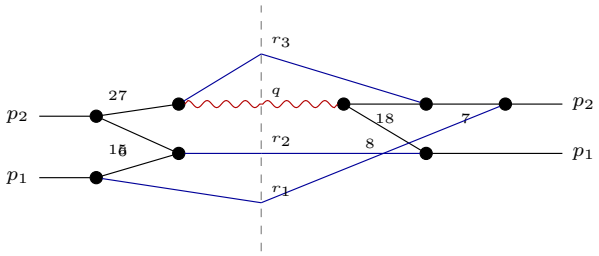
FAM[169] (topo169)
left: 6, 13, 22; right: 7, 16, 27; numerators: $12 = (r_1 + r_2)^2$



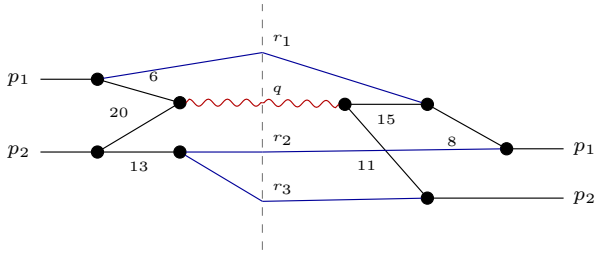
FAM[175] (topo175)
left: 6, 13, 22; right: 7, 16, 27; numerators: $15 = (p_1 + r_1 + r_2)^2$



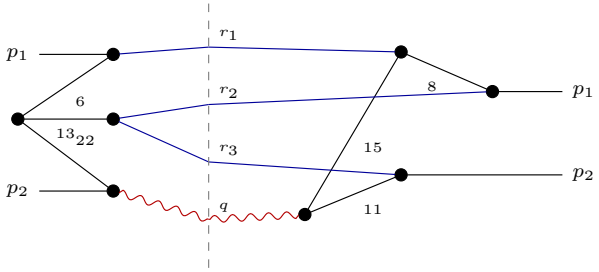
FAM[181] (topo181)
left: 6, 15, 27; right: 7, 16, 23; numerators: $18 = (p_2 + r_1 + r_3)^2$



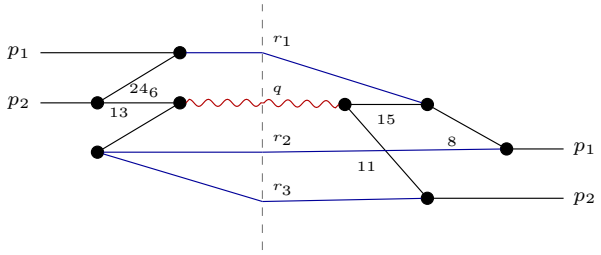
FAM[187] (topo187)
left: 6, 15, 27; right: 7, 8, 18; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



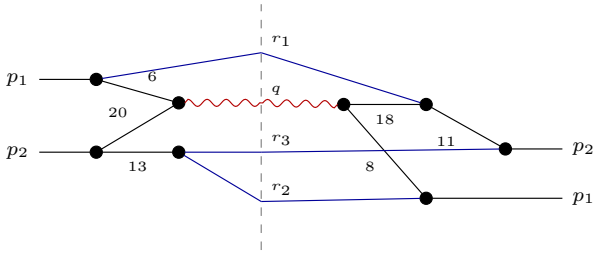
FAM[193] (topo193)
left: 6, 13, 20; right: 8, 11, 15; numerators: $21 = (r_1 + r_2 + r_3)^2$



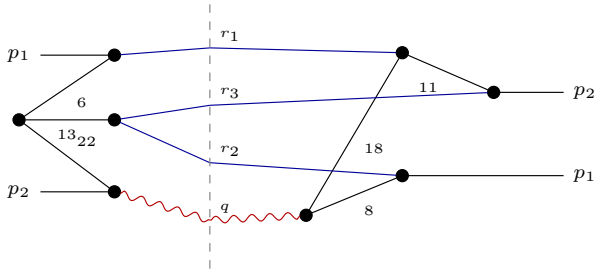
FAM[199] (topo199)
left: 6, 13, 22; right: 8, 11, 15; numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



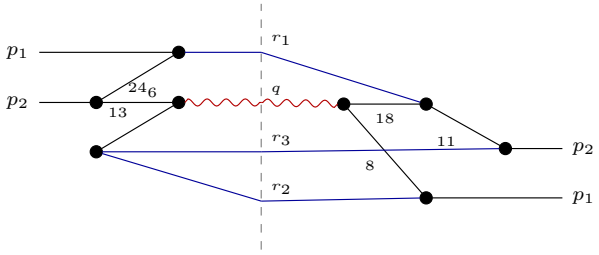
FAM[205] (topo205)
left: 6, 13, 24; right: 8, 11, 15; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



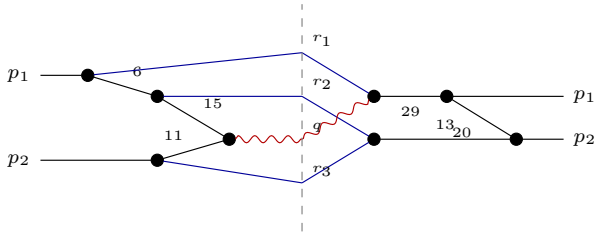
FAM[211] (topo211)
left: 6, 13, 20; right: 8, 11, 18; numerators: $19 = (p_1 + r_2 + r_3)^2$



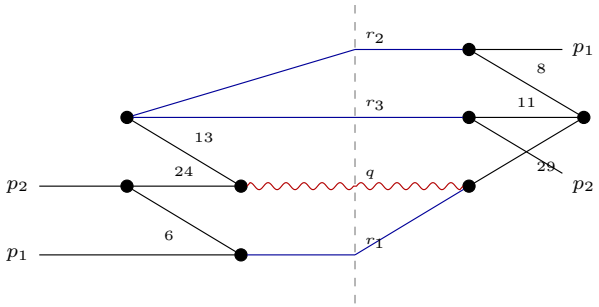
FAM[217] (topo217)
left: 6, 13, 22; right: 8, 11, 18; numerators: $17 = (p_1 + r_1 + r_3)^2$



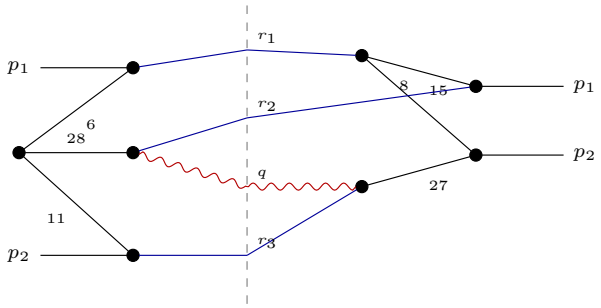
FAM[223] (topo223)
left: 6, 13, 24; right: 8, 11, 18; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



FAM[229] (topo229)
left: 6, 11, 15; right: 13, 20, 29; numerators: $8 = (p_1 + r_2)^2$

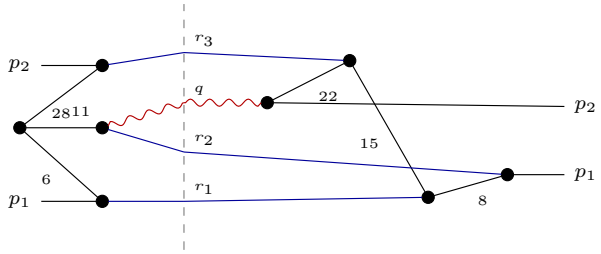


FAM[235] (topo235)
left: 6, 13, 24; right: 8, 11, 29; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



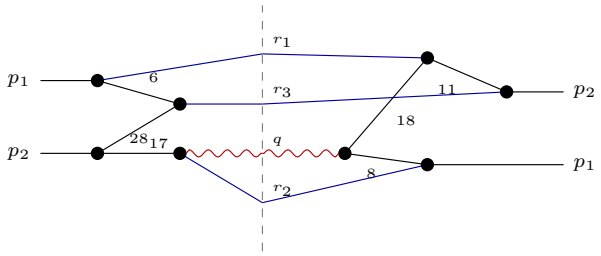
FAM[241] (topo241)

left: 6, 11, 28; right: 8, 15, 27; numerators: $17 = (p_1 + r_1 + r_3)^2$



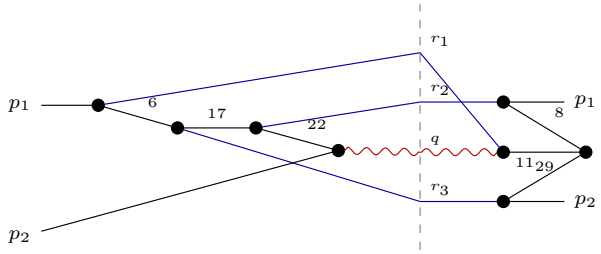
FAM[247] (topo247)

left: 6, 11, 28; right: 8, 15, 22; numerators: $24 = (p_1 + p_2 + r_1)^2$



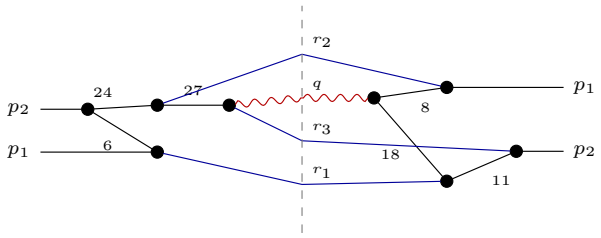
FAM[253] (topo253)

left: 6, 17, 28; right: 8, 11, 18; numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



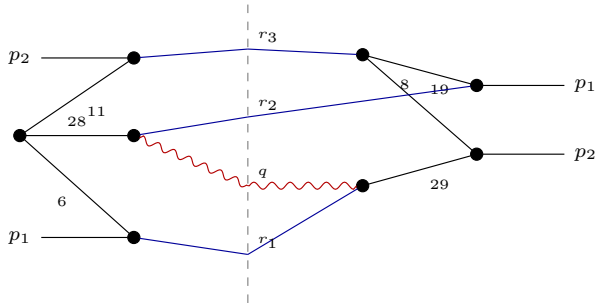
FAM[259] (topo259)

left: 6, 17, 22; right: 8, 11, 29; numerators: $25 = (p_1 + p_2 + r_2)^2$



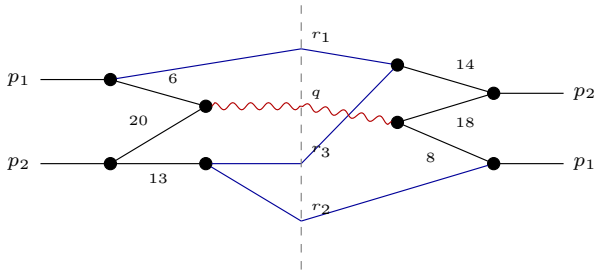
FAM[265] (topo265)

left: 6, 24, 27; right: 8, 11, 18; numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



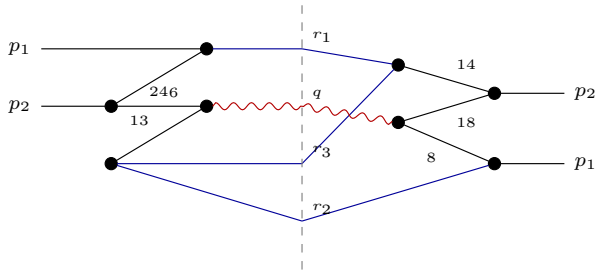
FAM[271] (topo271)

left: 6, 11, 28; right: 8, 19, 29; numerators: $24 = (p_1 + p_2 + r_1)^2$



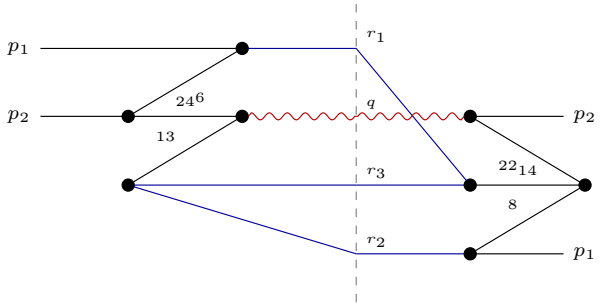
FAM[277] (topo277)

left: 6, 13, 20; right: 8, 14, 18; numerators: $21 = (r_1 + r_2 + r_3)^2$



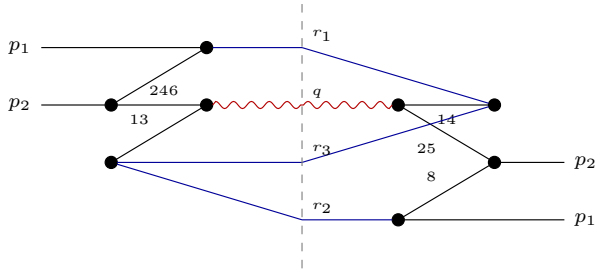
FAM[280] (topo280)

left: 6, 13, 24; right: 8, 14, 18; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



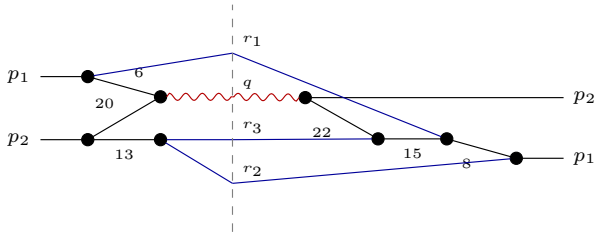
FAM[286] (topo286)

left: 6, 13, 24; right: 8, 14, 22; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$

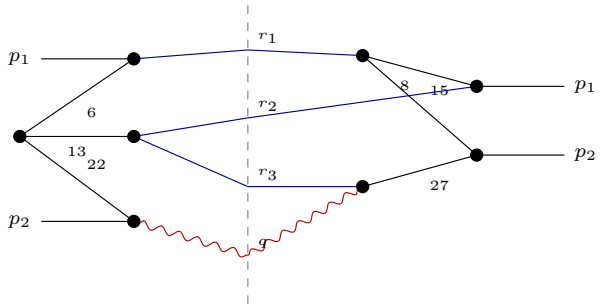


FAM[292] (topo292)

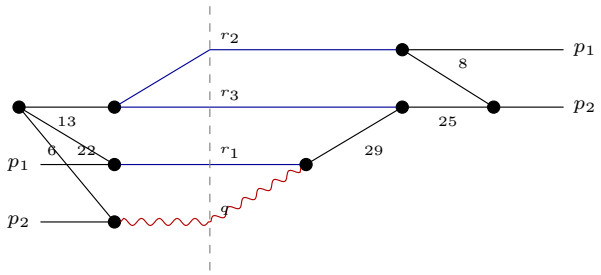
left: 6, 13, 24; right: 8, 14, 25; numerators: $17 = (p_1 + r_1 + r_3)^2$



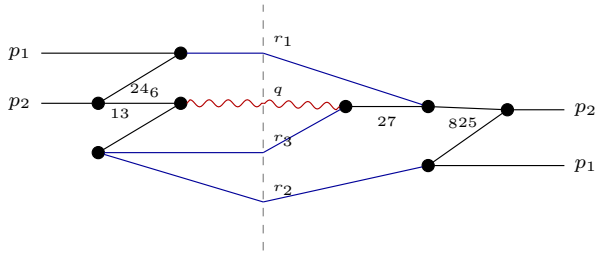
FAM[298] (topo298)
left: 6, 13, 20; right: 8, 15, 22; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



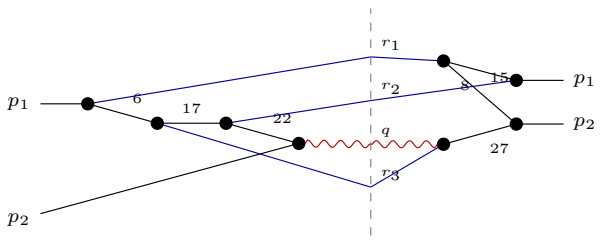
FAM[304] (topo304)
left: 6, 13, 22; right: 8, 15, 27; numerators: $24 = (p_1 + p_2 + r_1)^2$



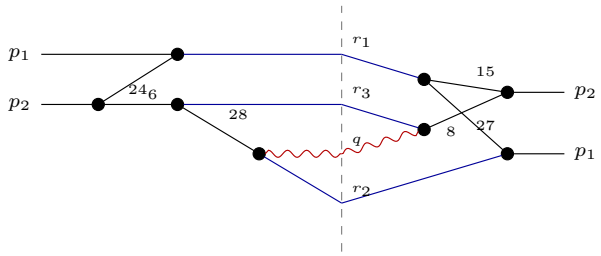
FAM[310] (topo310)
left: 6, 13, 22; right: 8, 25, 29; numerators: $12 = (r_1 + r_2)^2$



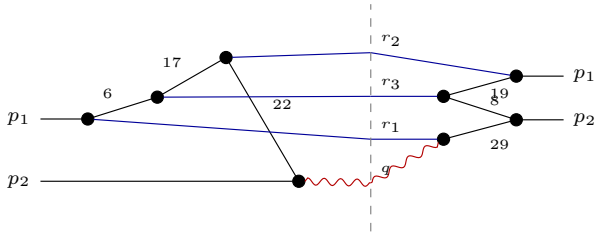
FAM[316] (topo316)
left: 6, 13, 24; right: 8, 25, 27; numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



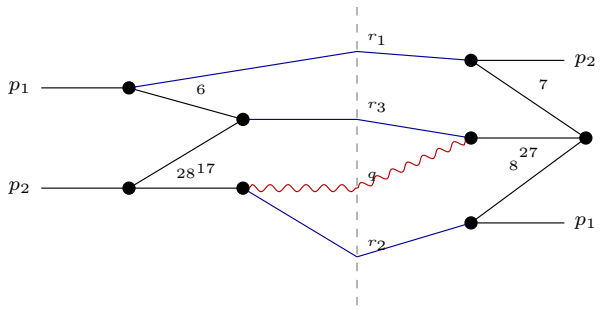
FAM[322] (topo322)
left: 6, 17, 22; right: 8, 15, 27; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



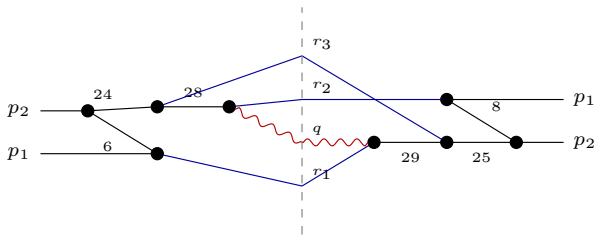
FAM[328] (topo328)
left: 6, 24, 28; right: 8, 15, 27; numerators: $10 = (p_1 + r_3)^2$



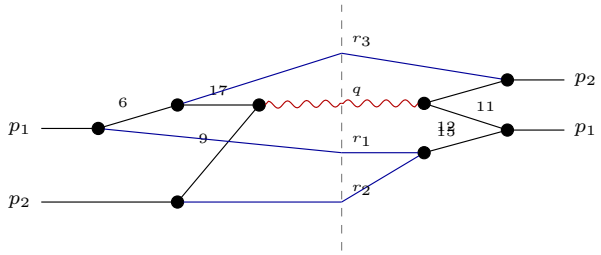
FAM[334] (topo334)
left: 6, 17, 22; right: 8, 19, 29; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



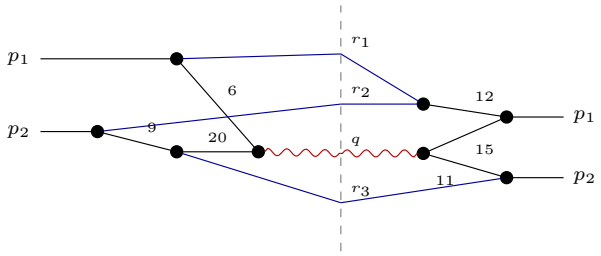
FAM[337] (topo337)
left: 6, 17, 28; right: 7, 8, 27; numerators: $25 = (p_1 + p_2 + r_2)^2$



FAM[343] (topo343)
left: 6, 24, 28; right: 8, 25, 29; numerators: $10 = (p_1 + r_3)^2$

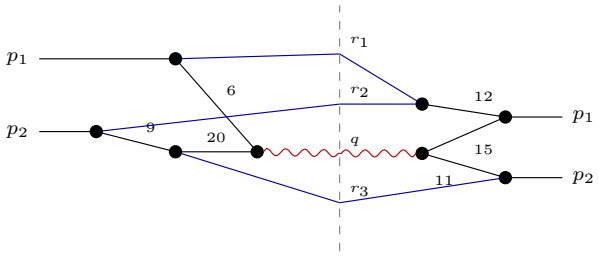


FAM[346] (topo346)
left: 6, 9, 17; right: 11, 12, 15; numerators: $16 = (p_2 + r_1 + r_2)^2$



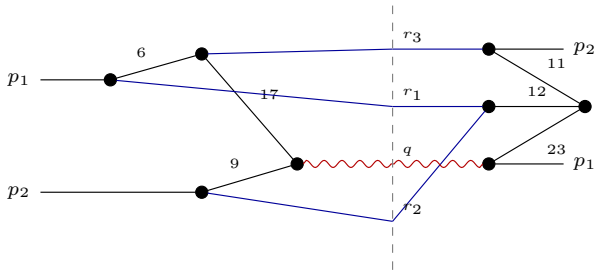
FAM[352] (topo352)

left: 6, 9, 20; right: 11, 12, 15; numerators: $21 = (r_1 + r_2 + r_3)^2$



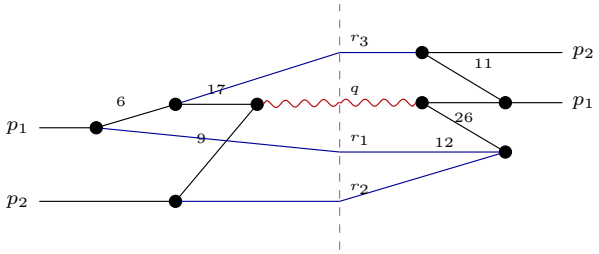
FAM[358] (topo358)

left: 6, 9, 20; right: 11, 12, 15; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



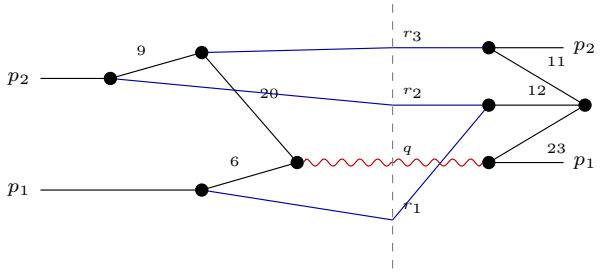
FAM[364] (topo364)

left: 6, 9, 17; right: 11, 12, 23; numerators: $18 = (p_2 + r_1 + r_3)^2$



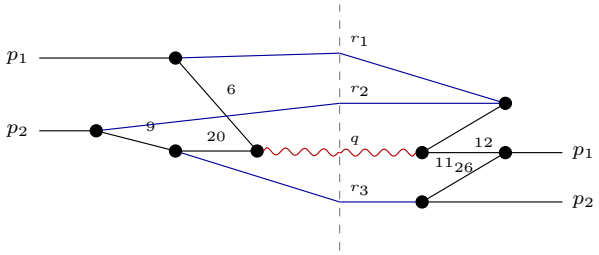
FAM[370] (topo370)

left: 6, 9, 17; right: 11, 12, 26; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



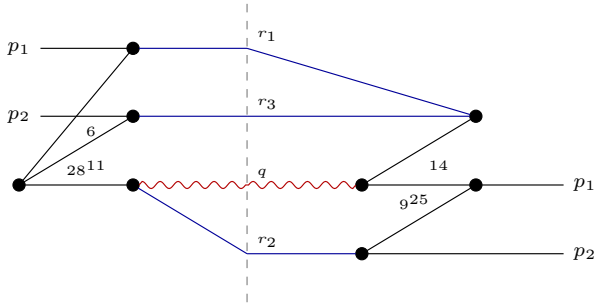
FAM[376] (topo376)

left: 6, 9, 20; right: 11, 12, 23; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



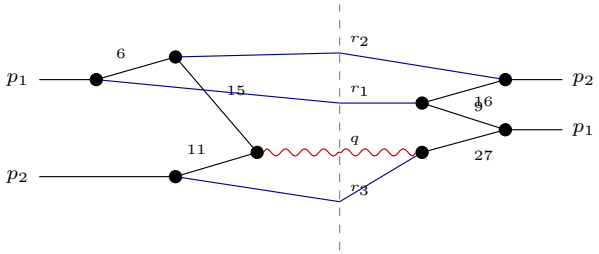
FAM[382] (topo382)

left: 6, 9, 20; right: 11, 12, 26; numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



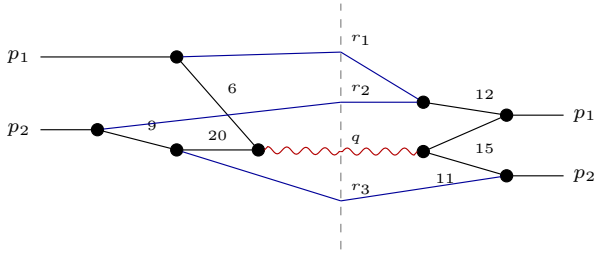
FAM[388] (topo388)

left: 6, 11, 28; right: 9, 14, 25; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



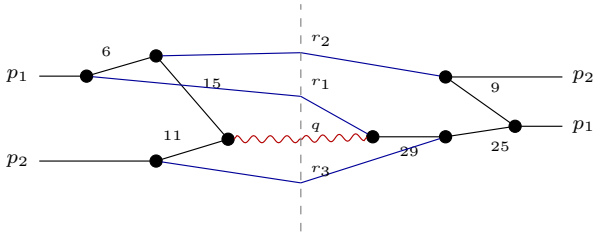
FAM[394] (topo394)

left: 6, 11, 15; right: 9, 16, 27; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



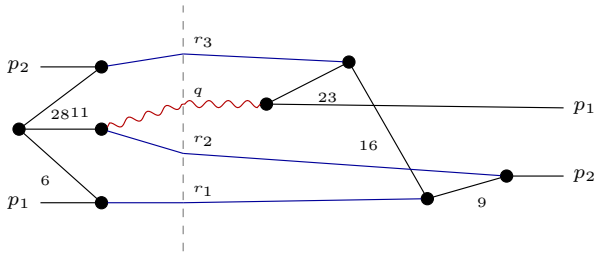
FAM[400] (topo400)

left: 6, 9, 20; right: 11, 12, 15; numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$

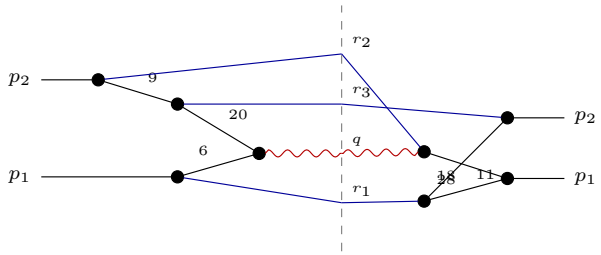


FAM[406] (topo406)

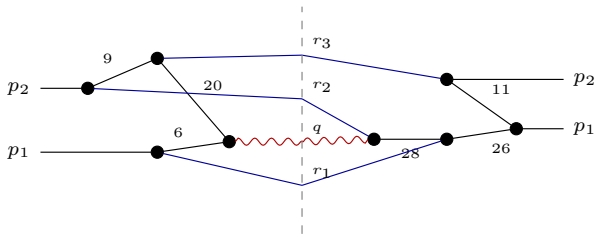
left: 6, 11, 15; right: 9, 25, 29; numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



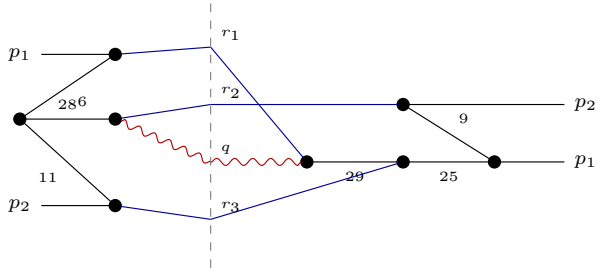
FAM[412] (topo412)
left: 6, 11, 28; right: 9, 16, 23; numerators: $26 = (p_1 + p_2 + r_3)^2$



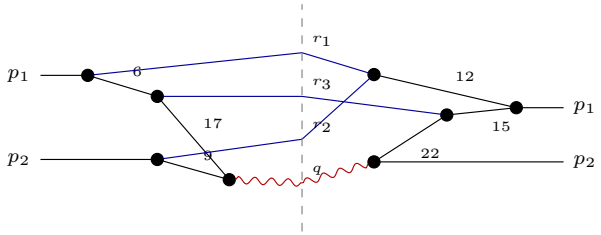
FAM[418] (topo418)
left: 6, 9, 20; right: 11, 18, 28; numerators: $14 = (r_1 + r_3)^2$



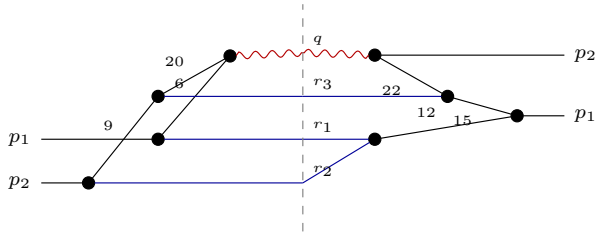
FAM[424] (topo424)
left: 6, 9, 20; right: 11, 26, 28; numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



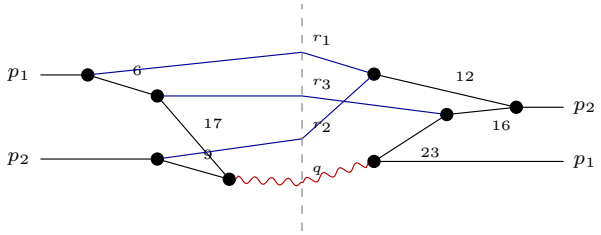
FAM[430] (topo430)
left: 6, 11, 28; right: 9, 25, 29; numerators: $17 = (p_1 + r_1 + r_3)^2$



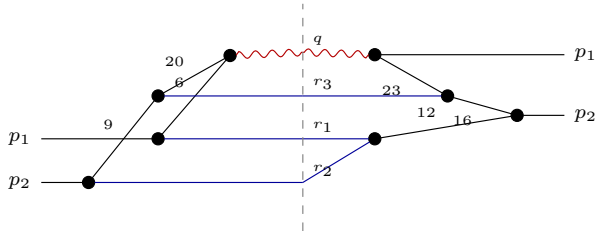
FAM[436] (topo436)
left: 6, 9, 17; right: 12, 15, 22; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



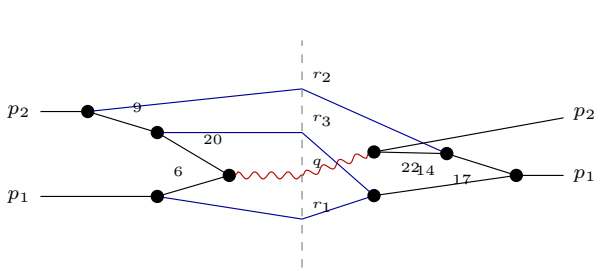
FAM[442] (topo442)
left: 6, 9, 20; right: 12, 15, 22; numerators: $16 = (p_2 + r_1 + r_2)^2$



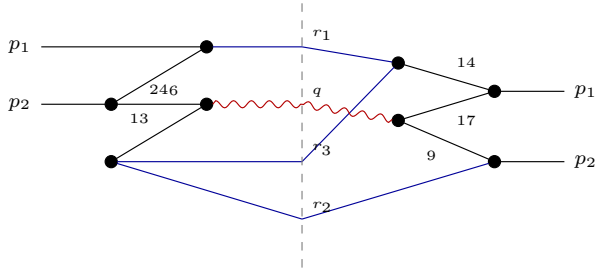
FAM[448] (topo448)
left: 6, 9, 17; right: 12, 16, 23; numerators: $15 = (p_1 + r_1 + r_2)^2$



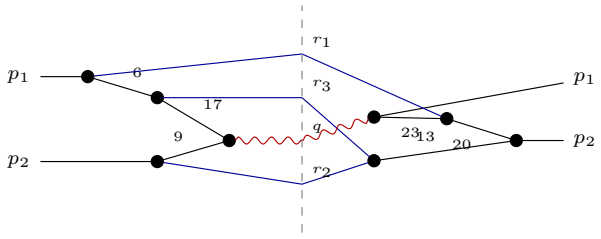
FAM[454] (topo454)
left: 6, 9, 20; right: 12, 16, 23; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



FAM[460] (topo460)
left: 6, 9, 20; right: 14, 17, 22; numerators: $13 = (r_2 + r_3)^2$

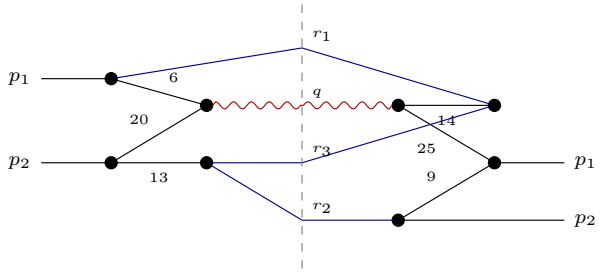


FAM[466] (topo466)
left: 6, 13, 24; right: 9, 14, 17; numerators: $21 = (r_1 + r_2 + r_3)^2$



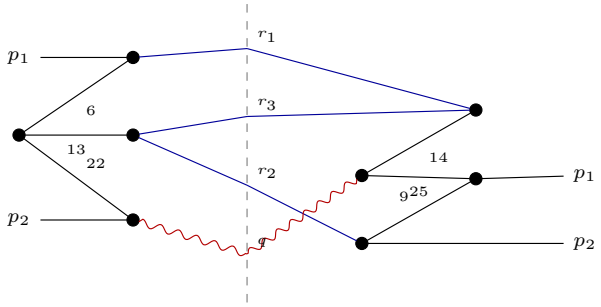
FAM[472] (topo472)

left: 6, 9, 17; right: 13, 20, 23; numerators: $14 = (r_1 + r_3)^2$



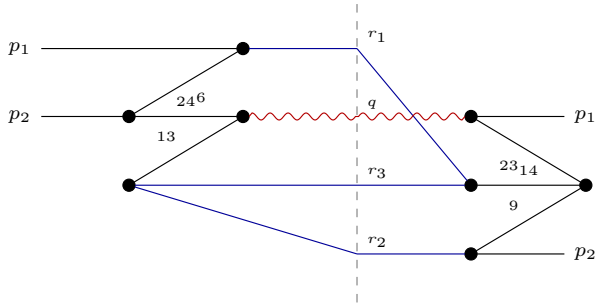
FAM[478] (topo478)

left: 6, 13, 20; right: 9, 14, 25; numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



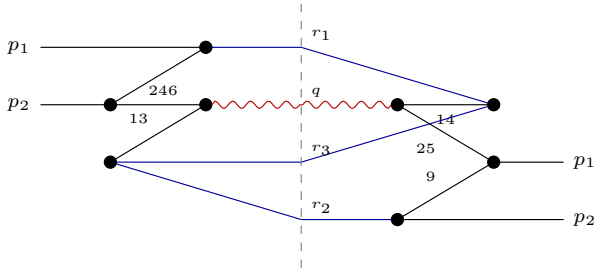
FAM[484] (topo484)

left: 6, 13, 22; right: 9, 14, 25; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



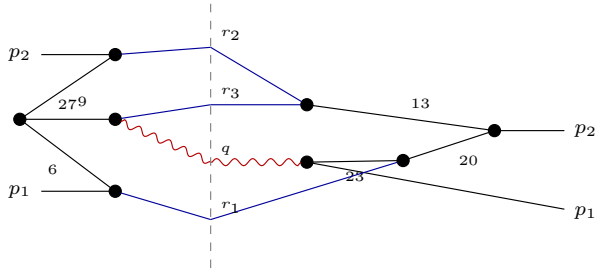
FAM[490] (topo490)

left: 6, 13, 24; right: 9, 14, 23; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



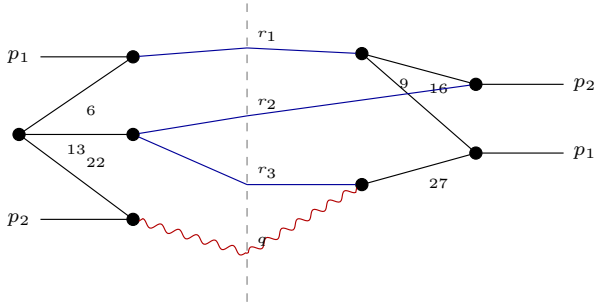
FAM[496] (topo496)

left: 6, 13, 24; right: 9, 14, 25; numerators: $21 = (r_1 + r_2 + r_3)^2$



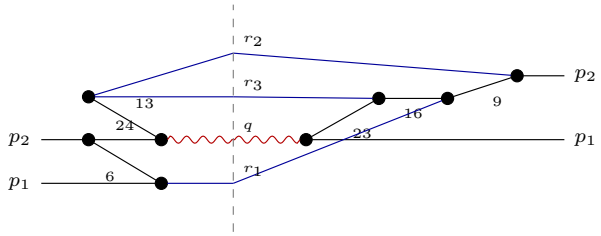
FAM[502] (topo502)

left: 6, 9, 27; right: 13, 20, 23; numerators: $16 = (p_2 + r_1 + r_2)^2$



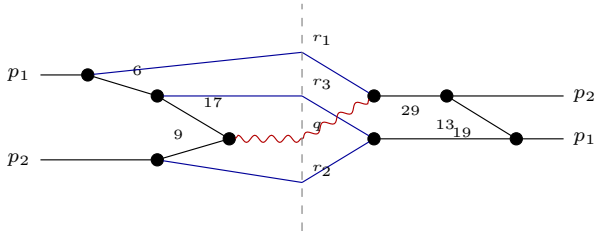
FAM[508] (topo508)

left: 6, 13, 22; right: 9, 16, 27; numerators: $15 = (p_1 + r_1 + r_2)^2$



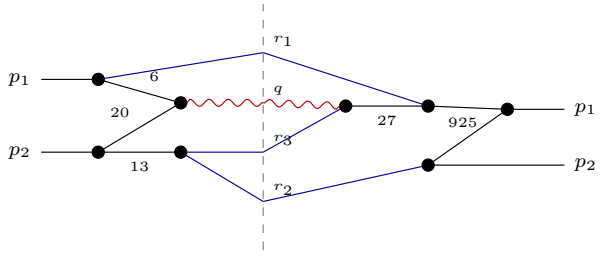
FAM[514] (topo514)

left: 6, 13, 24; right: 9, 16, 23; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



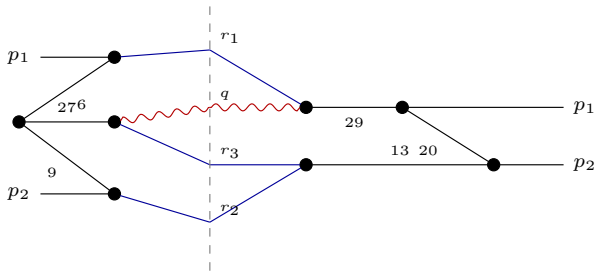
FAM[520] (topo520)

left: 6, 9, 17; right: 13, 19, 29; numerators: $14 = (r_1 + r_3)^2$

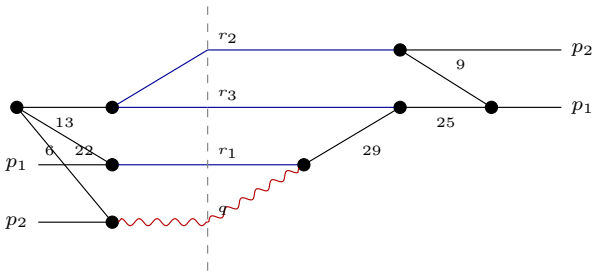


FAM[526] (topo526)

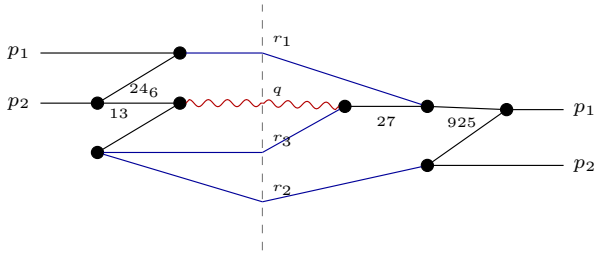
left: 6, 13, 20; right: 9, 25, 27; numerators: $21 = (r_1 + r_2 + r_3)^2$



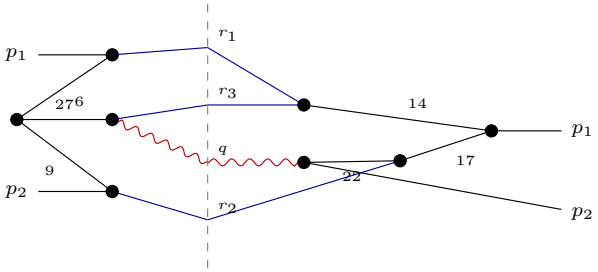
FAM[532] (topo532)
left: 6, 9, 27; right: 13, 20, 29; numerators: $16 = (p_2 + r_1 + r_2)^2$



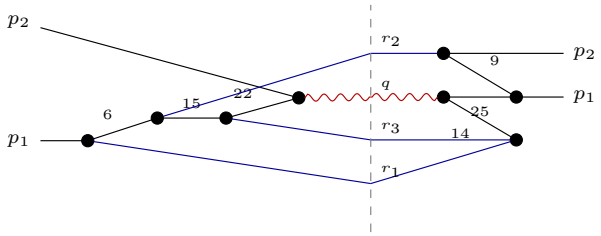
FAM[538] (topo538)
left: 6, 13, 22; right: 9, 25, 29; numerators: $17 = (p_1 + r_1 + r_3)^2$



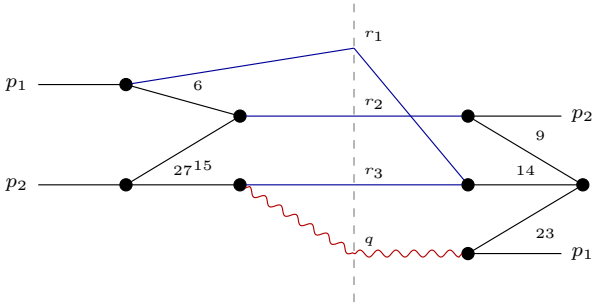
FAM[544] (topo544)
left: 6, 13, 24; right: 9, 25, 27; numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



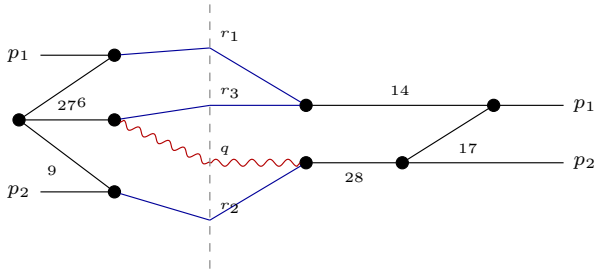
FAM[550] (topo550)
left: 6, 9, 27; right: 14, 17, 22; numerators: $15 = (p_1 + r_1 + r_2)^2$



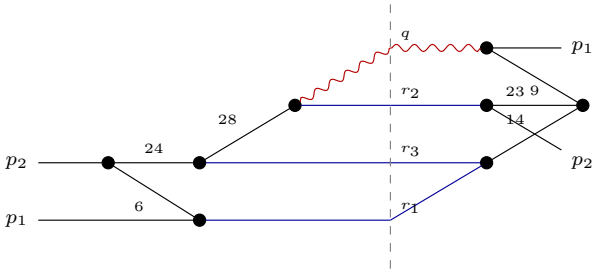
FAM[556] (topo556)
left: 6, 15, 22; right: 9, 14, 25; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



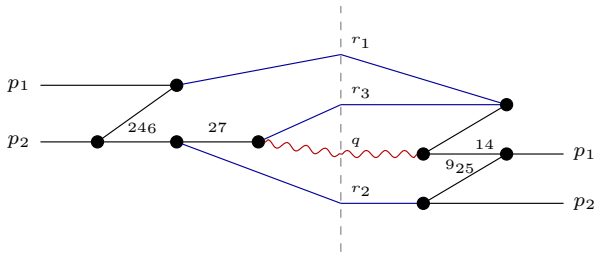
FAM[562] (topo562)
left: 6, 15, 27; right: 9, 14, 23; numerators: $20 = (p_2 + r_2 + r_3)^2$



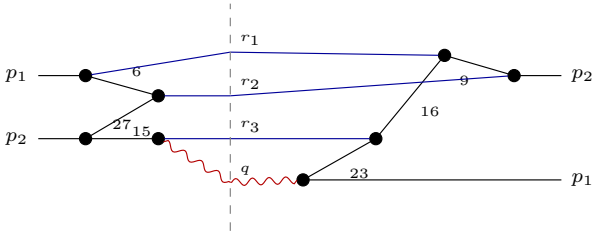
FAM[568] (topo568)
left: 6, 9, 27; right: 14, 17, 28; numerators: $24 = (p_1 + p_2 + r_1)^2$



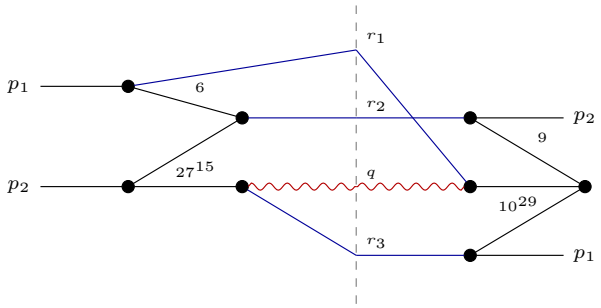
FAM[574] (topo574)
left: 6, 24, 28; right: 9, 14, 23; numerators: $20 = (p_2 + r_2 + r_3)^2$



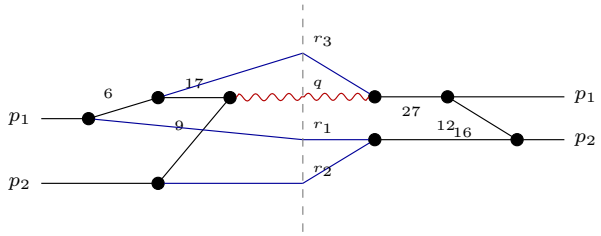
FAM[580] (topo580)
left: 6, 24, 27; right: 9, 14, 25; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



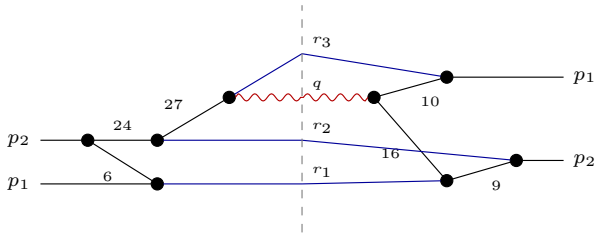
FAM[586] (topo586)
left: 6, 15, 27; right: 9, 16, 23; numerators: $13 = (r_2 + r_3)^2$



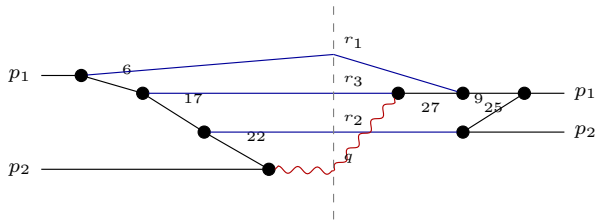
FAM[592] (topo592)
left: 6, 15, 27; right: 9, 10, 29; numerators: $25 = (p_1 + p_2 + r_2)^2$



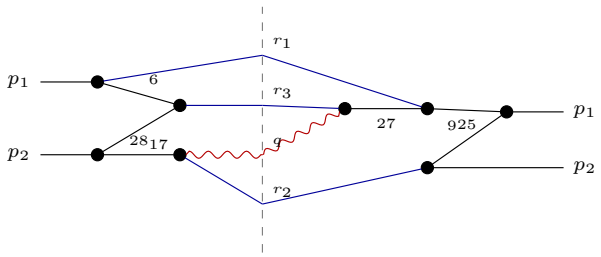
FAM[598] (topo598)
left: 6, 9, 17; right: 12, 16, 27; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



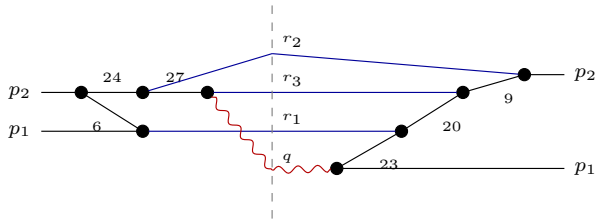
FAM[604] (topo604)
left: 6, 24, 27; right: 9, 10, 16; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



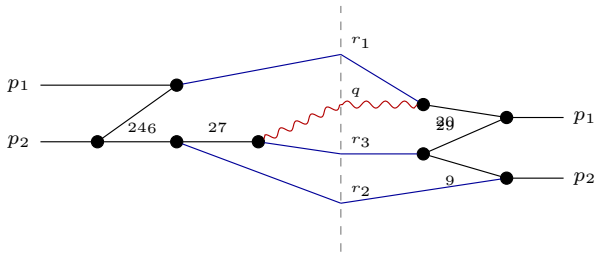
FAM[610] (topo610)
left: 6, 17, 22; right: 9, 25, 27; numerators: $7 = (p_2 + r_1)^2$



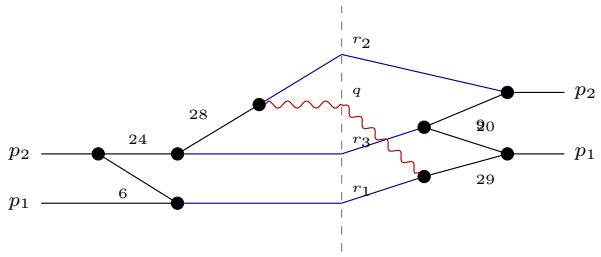
FAM[616] (topo616)
left: 6, 17, 28; right: 9, 25, 27; numerators: $7 = (p_2 + r_1)^2$



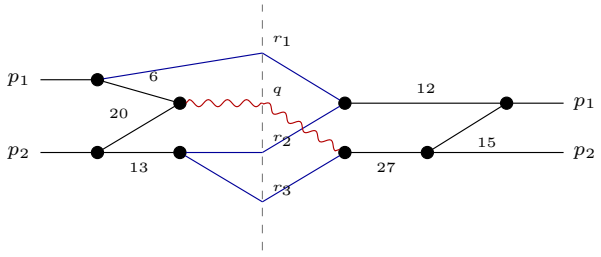
FAM[622] (topo622)
left: 6, 24, 27; right: 9, 20, 23; numerators: $8 = (p_1 + r_2)^2$



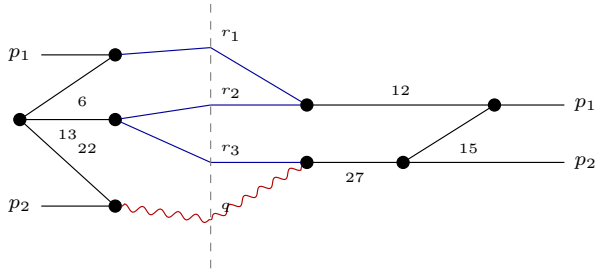
FAM[628] (topo628)
left: 6, 24, 27; right: 9, 20, 29; numerators: $8 = (p_1 + r_2)^2$



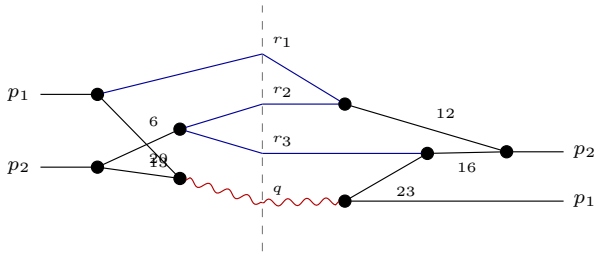
FAM[634] (topo634)
left: 6, 24, 28; right: 9, 20, 29; numerators: $25 = (p_1 + p_2 + r_2)^2$



FAM[640] (topo640)
left: 6, 13, 20; right: 12, 15, 27; numerators: $21 = (r_1 + r_2 + r_3)^2$

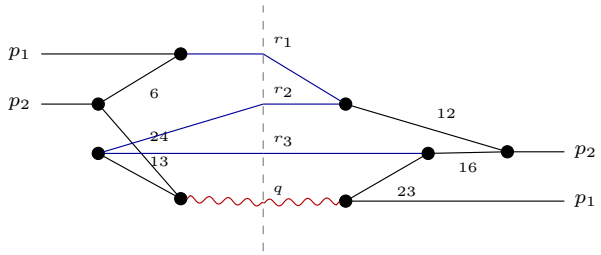


FAM[646] (topo646)
left: 6, 13, 22; right: 12, 15, 27; numerators: $24 = (p_1 + p_2 + r_1)^2$



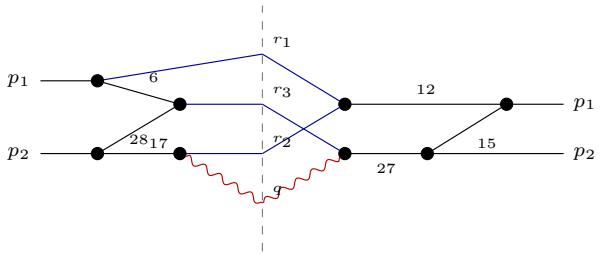
FAM[652] (topo652)

left: 6, 13, 20; right: 12, 16, 23; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



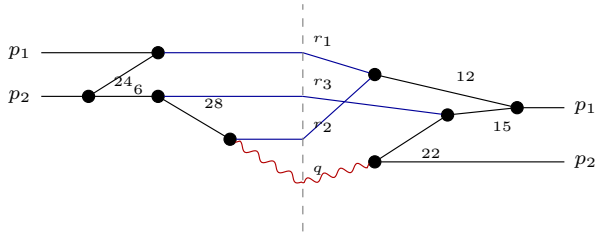
FAM[658] (topo658)

left: 6, 13, 24; right: 12, 16, 23; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



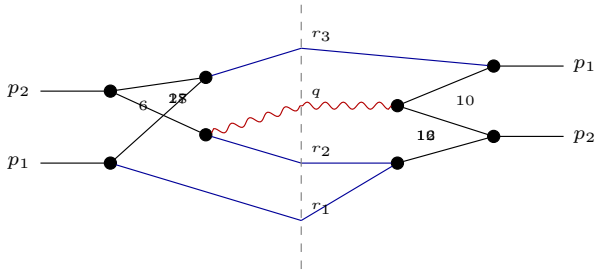
FAM[664] (topo664)

left: 6, 17, 28; right: 12, 15, 27; numerators: $26 = (p_1 + p_2 + r_3)^2$



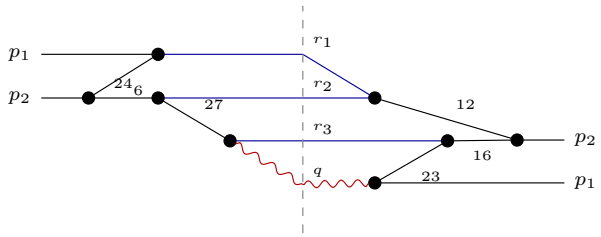
FAM[670] (topo670)

left: 6, 24, 28; right: 12, 15, 22; numerators: $26 = (p_1 + p_2 + r_3)^2$



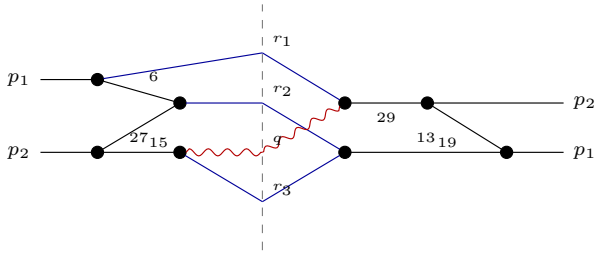
FAM[676] (topo676)

left: 6, 17, 28; right: 10, 12, 16; numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



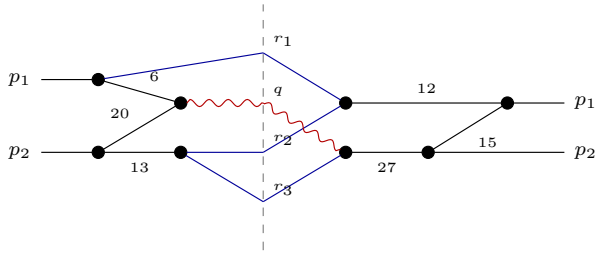
FAM[682] (topo682)

left: 6, 24, 27; right: 12, 16, 23; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



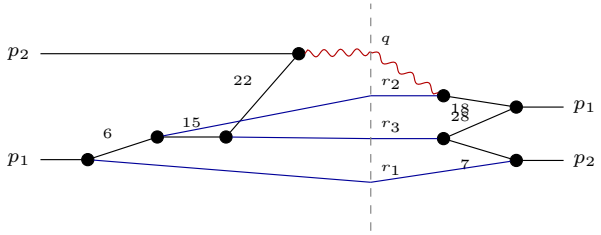
FAM[688] (topo688)

left: 6, 15, 27; right: 13, 19, 29; numerators: $12 = (r_1 + r_2)^2$



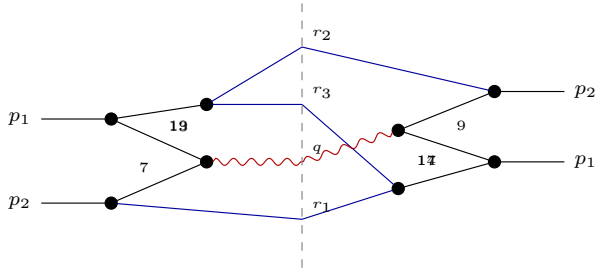
FAM[694] (topo694)

left: 6, 13, 20; right: 12, 15, 27; numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



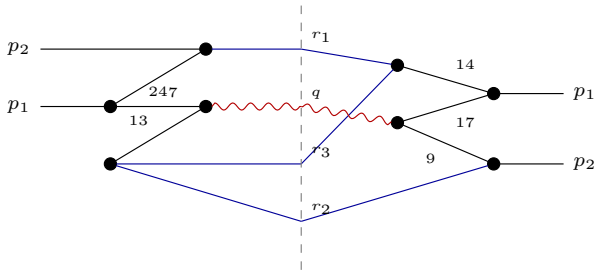
FAM[700] (topo700)

left: 6, 15, 22; right: 7, 18, 28; numerators: $14 = (r_1 + r_3)^2$

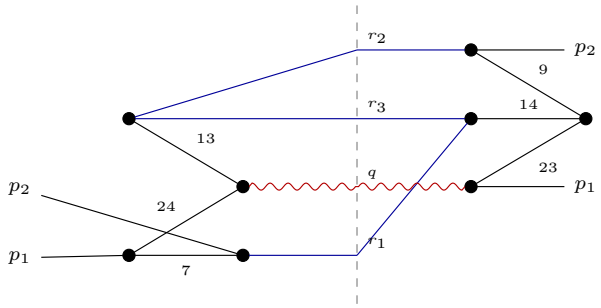


FAM[706] (topo706)

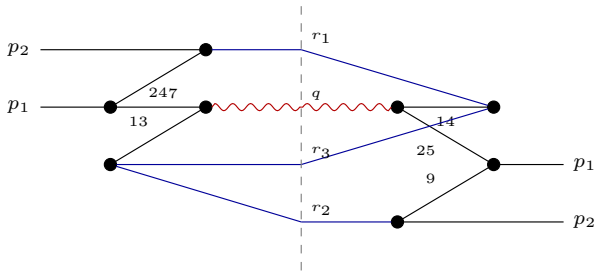
left: 7, 13, 19; right: 9, 14, 17; numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



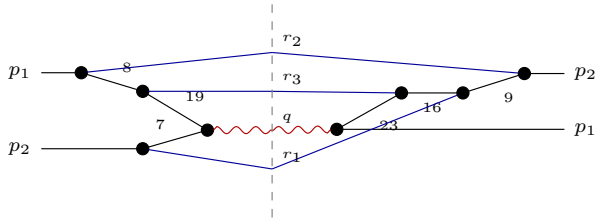
FAM[709] (topo709)
left: 7, 13, 24; right: 9, 14, 17; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



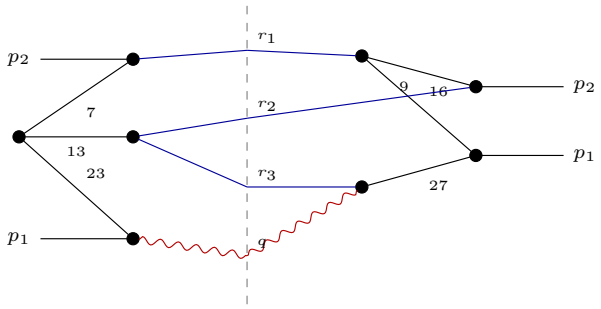
FAM[715] (topo715)
left: 7, 13, 24; right: 9, 14, 23; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



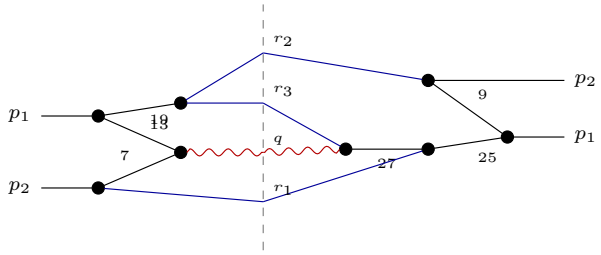
FAM[721] (topo721)
left: 7, 13, 24; right: 9, 14, 25; numerators: $18 = (p_2 + r_1 + r_3)^2$



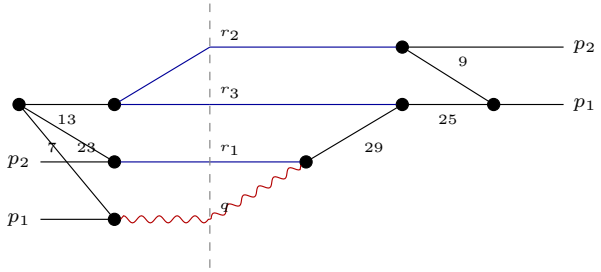
FAM[727] (topo727)
left: 7, 8, 19; right: 9, 16, 23; numerators: $13 = (r_2 + r_3)^2$



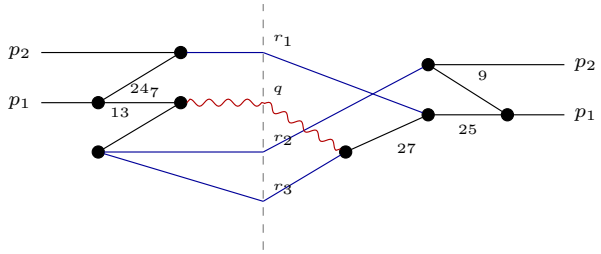
FAM[733] (topo733)
left: 7, 13, 23; right: 9, 16, 27; numerators: $24 = (p_1 + p_2 + r_1)^2$



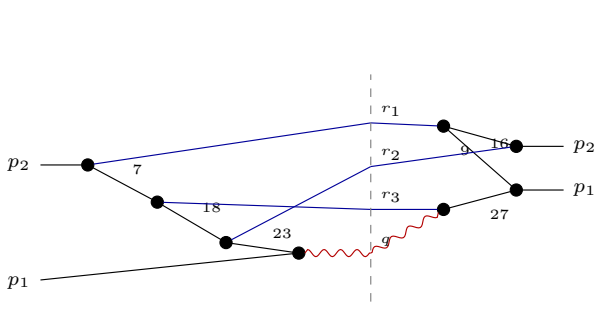
FAM[739] (topo739)
left: 7, 13, 19; right: 9, 25, 27; numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



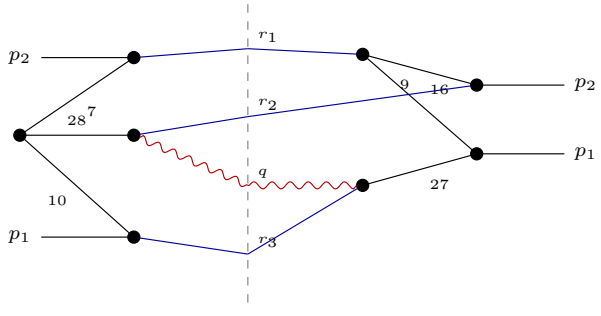
FAM[745] (topo745)
left: 7, 13, 23; right: 9, 25, 29; numerators: $12 = (r_1 + r_2)^2$



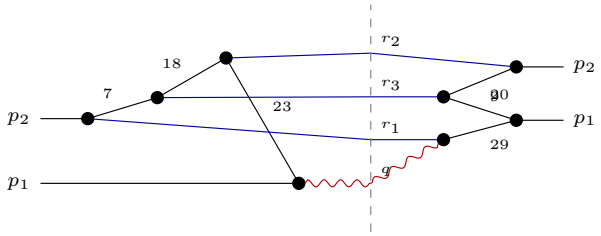
FAM[751] (topo751)
left: 7, 13, 24; right: 9, 25, 27; numerators: $29 = (p_1 + p_2 + r_2 + r_3)^2$



FAM[757] (topo757)
left: 7, 18, 23; right: 9, 16, 27; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$

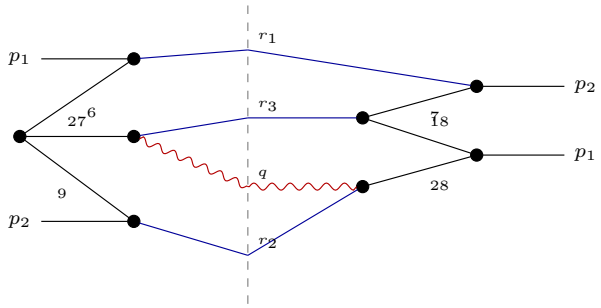


FAM[763] (topo763)
left: 7, 10, 28; right: 9, 16, 27; numerators: $24 = (p_1 + p_2 + r_1)^2$



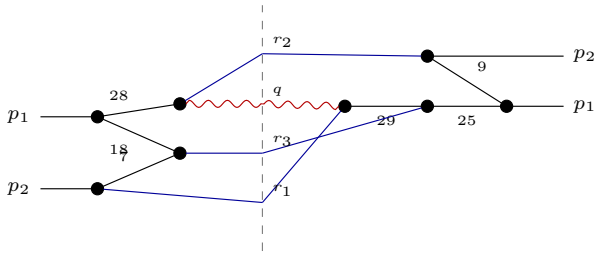
FAM[769] (topo769)

left: 7, 18, 23; right: 9, 20, 29; numerators: $28 = (p_1 + p_2 + r_1 + r_3)^2$



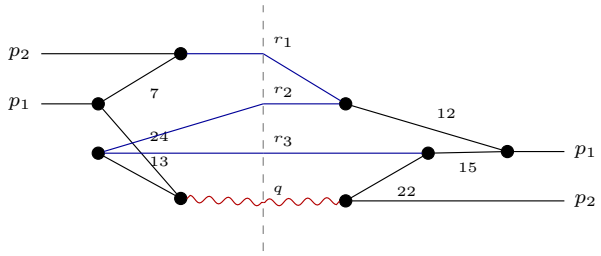
FAM[772] (topo772)

left: 6, 9, 27; right: 7, 18, 28; numerators: $25 = (p_1 + p_2 + r_2)^2$



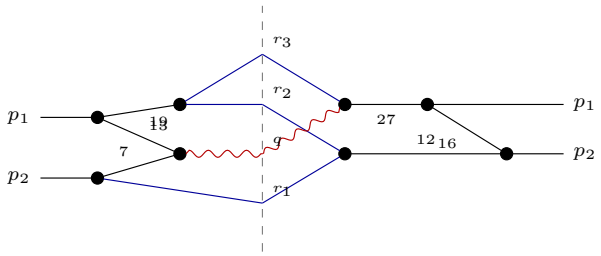
FAM[778] (topo778)

left: 7, 18, 28; right: 9, 25, 29; numerators: $24 = (p_1 + p_2 + r_1)^2$



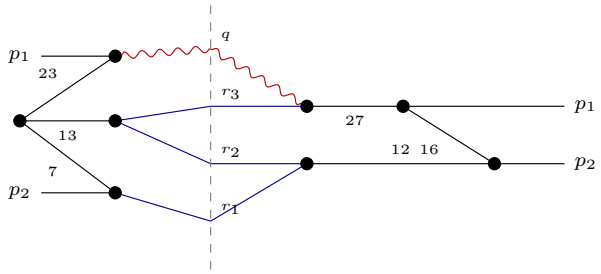
FAM[784] (topo784)

left: 7, 13, 24; right: 12, 15, 22; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



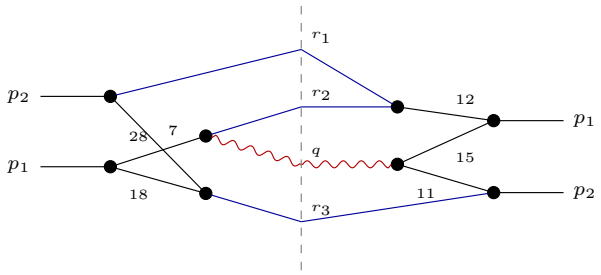
FAM[790] (topo790)

left: 7, 13, 19; right: 12, 16, 27; numerators: $21 = (r_1 + r_2 + r_3)^2$



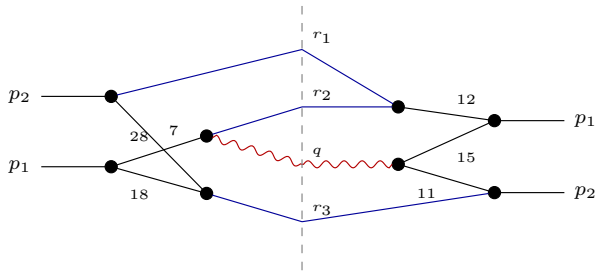
FAM[796] (topo796)

left: 7, 13, 23; right: 12, 16, 27; numerators: $24 = (p_1 + p_2 + r_1)^2$



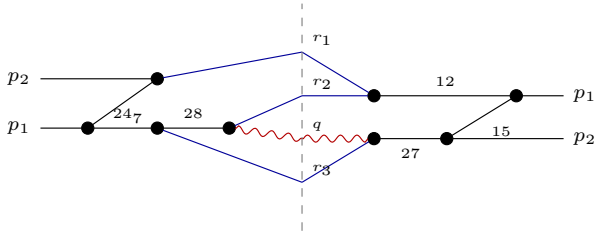
FAM[802] (topo802)

left: 7, 18, 28; right: 11, 12, 15; numerators: $22 = (p_1 + r_1 + r_2 + r_3)^2$



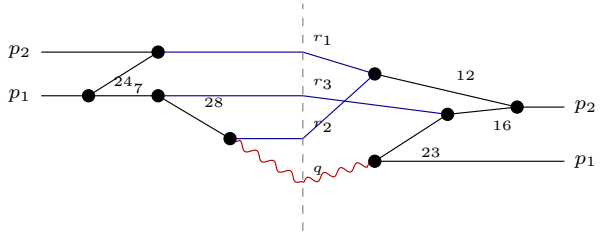
FAM[808] (topo808)

left: 7, 18, 28; right: 11, 12, 15; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



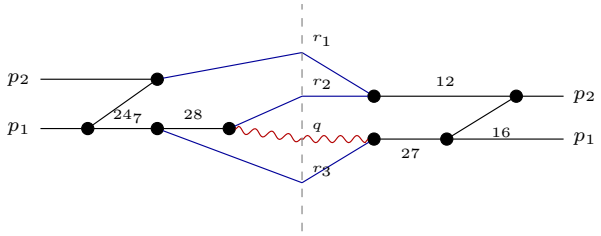
FAM[814] (topo814)

left: 7, 24, 28; right: 12, 15, 27; numerators: $21 = (r_1 + r_2 + r_3)^2$



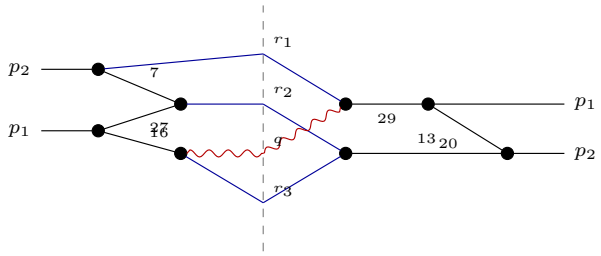
FAM[820] (topo820)

left: 7, 24, 28; right: 12, 16, 23; numerators: $26 = (p_1 + p_2 + r_3)^2$



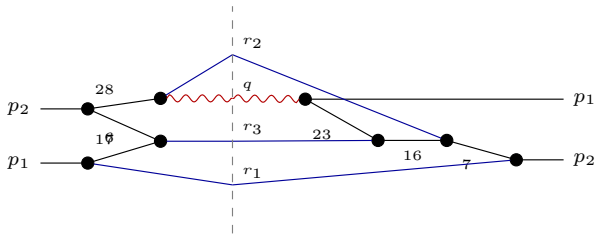
FAM[826] (topo826)

left: 7, 24, 28; right: 12, 16, 27; numerators: $26 = (p_1 + p_2 + r_3)^2$



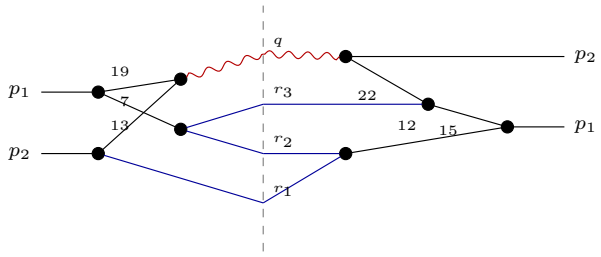
FAM[832] (topo832)

left: 7, 16, 27; right: 13, 20, 29; numerators: $12 = (r_1 + r_2)^2$



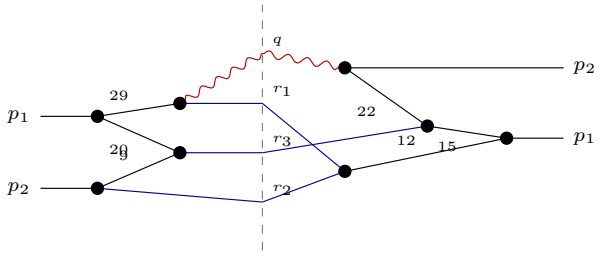
FAM[838] (topo838)

left: 6, 17, 28; right: 7, 16, 23; numerators: $14 = (r_1 + r_3)^2$



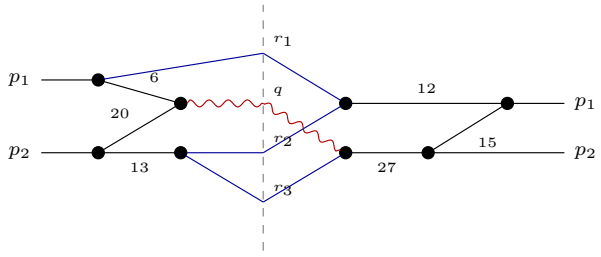
FAM[844] (topo844)

left: 7, 13, 19; right: 12, 15, 22; numerators: $27 = (p_1 + p_2 + r_1 + r_2)^2$



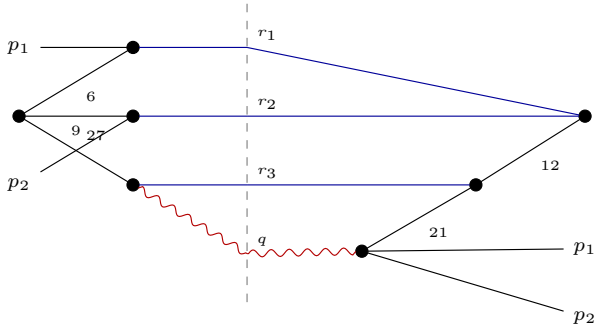
FAM[850] (topo850)

left: 9, 20, 29; right: 12, 15, 22; numerators: $13 = (r_2 + r_3)^2$



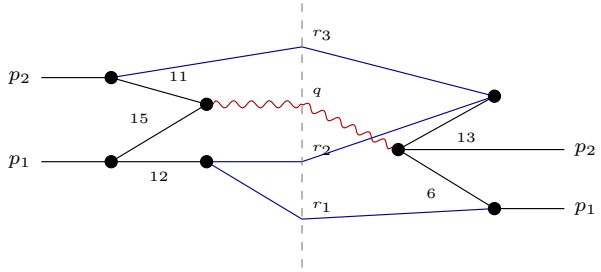
FAM[856] (topo856)

left: 6, 13, 20; right: 12, 15, 27; numerators: $23 = (p_2 + r_1 + r_2 + r_3)^2$



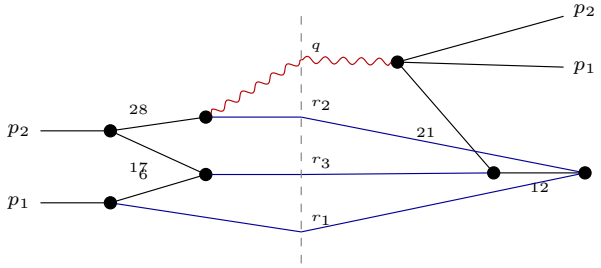
FAM[862] (topo862)

left: 6, 9, 27; right: 12, 21; numerators: $14 = (r_1 + r_3)^2$, $24 = (p_1 + p_2 + r_1)^2$



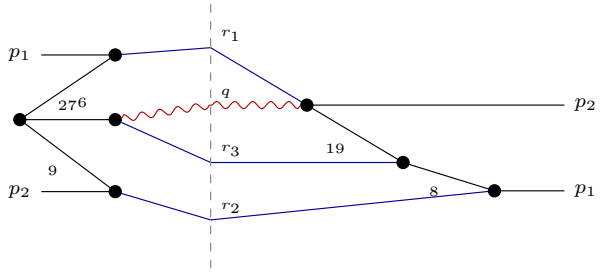
FAM[868] (topo868)

left: 11, 12, 15; right: 6, 13; numerators: $7 = (p_2 + r_1)^2$, $26 = (p_1 + p_2 + r_3)^2$



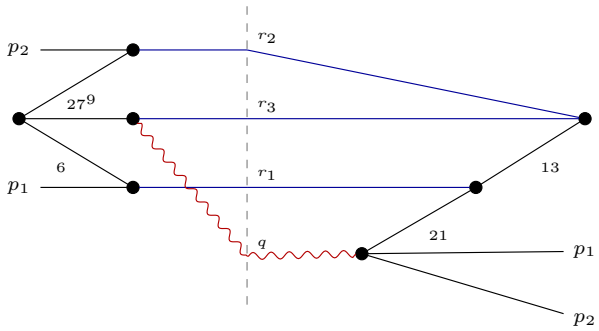
FAM[874] (topo874)

left: 6, 17, 28; right: 12, 21; numerators: $7 = (p_2 + r_1)^2$, $18 = (p_2 + r_1 + r_3)^2$

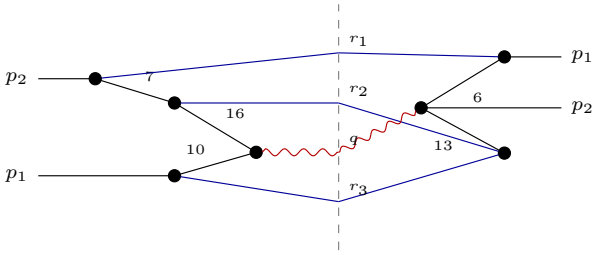


FAM[880] (topo880)

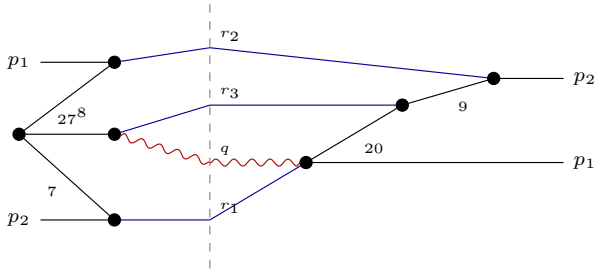
left: 6, 9, 27; right: 8, 19; numerators: $13 = (r_2 + r_3)^2$, $24 = (p_1 + p_2 + r_1)^2$



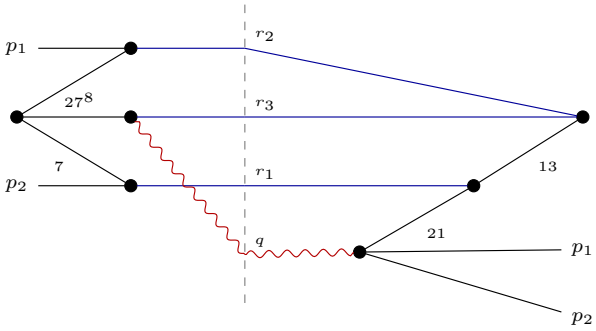
FAM[886] (topo886)
left: 6, 9, 27; right: 13, 21; numerators: $8 = (p_1 + r_2)^2$, $24 = (p_1 + p_2 + r_1)^2$



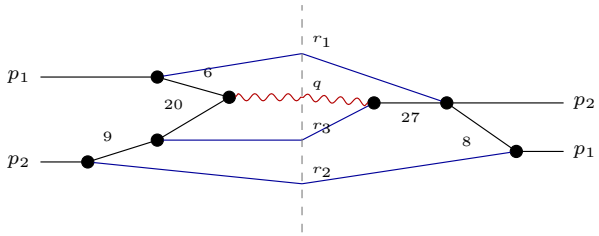
FAM[892] (topo892)
left: 7, 10, 16; right: 6, 13; numerators: $12 = (r_1 + r_2)^2$, $26 = (p_1 + p_2 + r_3)^2$



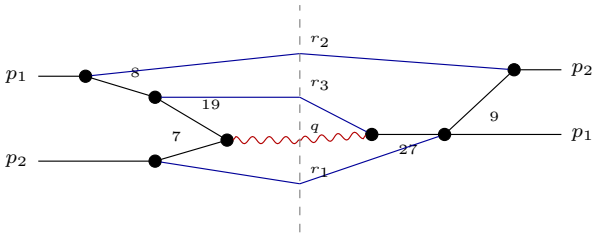
FAM[898] (topo898)
left: 7, 8, 27; right: 9, 20; numerators: $13 = (r_2 + r_3)^2$, $24 = (p_1 + p_2 + r_1)^2$



FAM[904] (topo904)
left: 7, 8, 27; right: 13, 21; numerators: $9 = (p_2 + r_2)^2$, $24 = (p_1 + p_2 + r_1)^2$



FAM[910] (topo910)
left: 6, 9, 20; right: 8, 27; numerators: $13 = (r_2 + r_3)^2$, $24 = (p_1 + p_2 + r_1)^2$



FAM[916] (topo916)
left: 7, 8, 19; right: 9, 27; numerators: $13 = (r_2 + r_3)^2$, $24 = (p_1 + p_2 + r_1)^2$